课程号：CS1017F，开课学院：计算机

考试试卷：√A卷、B卷（请在选定项上打√）

考试形式：√闭、开卷（请在选定项上打√），允许带________入场

考试日期：2025 年 4 月 27 日 16:15-18:00，考试时间：105 分钟

诚信考试，沉着应考，杜绝违纪。

考生姓名：________ 班号：________

1. (20 points) Determine whether the following statements are true or false. If it i[s] true write a √ otherwise a × in the blank before the statement.

( ) (1) The proposition "2·3=8 if and only if 3+2=4" is True.

( ) (2) The following system specifications are consistent.

Whenever the system software is being upgraded, users cannot access the file syst[em].

If users can access the file system, then they can save new files.

If users cannot save new files, then the system software is not being upgraded.

( ) (3) Suppose $A=\{a,b,c\}$, then $\{a\}\subset P(A)$. ($P(S)$ is the power set of S.)

( ) (4) $\overline{\overline{A}\cup\overline{B}}\cup\overline{A}=\overline{A}$

( ) (5) $A\oplus A=A$ ($\oplus$ is the symmetric difference of two sets.)

( ) (6) Function f: $N\to N$, defined as $f(x)=\lceil x\rceil+1$, is onto.

( ) (7) $(Z\times Z\times Z)-(R^+\times R^+\times R^+)$ is an uncountable set.

( ) (8) Let $F=\{f\mid f: N\to\{0,1\}\}$, where N is the set of natural numbers, then F is coun[t]
infinite.

( ) (9) $\sum_{j=1}^{n}(j^3+j)$ is $O(n^4)$.

( ) (10) In a group of five people (where any two people are either friends or enemie[s]) there are either three mutual friends or three mutual enemies.

2. (30 points) Fill in the blanks.

(1) (2pts) Let $F(A)$ be the predicate "A is a finite set" and $S(A, B)$ be the predicate "A is in B". Suppose the domain consists of all sets. Translate the statement into symbols: ***Every subset of a finite set is finite.***

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(2) (2pts) The prenex DNF of $(\exists x P(x) \land \forall x Q(x)) \rightarrow \exists x(\neg P(x) \rightarrow Q(x))$ is: ______

(3) (2pts) The power set of $\{\varnothing, \{\varnothing\}\}$ is ______

(4) (2pts) Suppose $g: A \rightarrow B$ and $f: B \rightarrow C$ where $A = \{1,2,3,4\}$, $B = \{a,b,c\}$, $C = \{3,7,10\}$, and f and g are defined by $g = \{(1,b),(2,a),(3,c),(4,b)\}$ and $f = \{(a,10),(b,7),(c,3)\}$. $f \circ g =$ ______

(5) (2pts) How many different truth tables of compound propositions are there that involve the propositional variables p and q? ______

(6) (2pts) Arrange the functions $(1.5)^n, n^{100}, (\log n)^3, \sqrt{n}\log n, 10^n, (n!)^2$, and $n^{99} + n^{98}$ in a list so that **each function is big-O of the next function.**

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(7) (2pts) Suppose $|A| = 6$ and $|B| = 12$. The number of 1-1 functions $f: A \rightarrow B$ is ______

If $|A| = 12$ and $|B| = 6$, then the number of 1-1 functions $f: A \rightarrow B$ is ______

(8) (6pts) You have 19 cards and 10 envelopes (labeled 1, 2, ..., 10). In how many ways can you put the 19 cards into the envelopes if

(a) the cards are distinct. ______

(b) the cards are identical. ______

(c) the cards are identical and no envelope can be left empty. ______

(9) (2pts) The number of permutations of the letters in the word ***TATTERED*** is

______

(10) (2pts) The next larger permutation in lexicographic order after 31528764 is

______

(11) (2pts) The next **four** larger 4-combinations of the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ after $\{2, 3, 4, 8\}$ is ______

(12) (2pts) The number of bit strings of length n, where $n \geqslant 4$, containing exactly two occurrences of 01 is ______

(13) (2pts) The number of ordered pairs $(\boldsymbol{A}, \boldsymbol{B})$ such that $\boldsymbol{A}$ and $\boldsymbol{B}$ are subsets of $\boldsymbol{S}$ with $\boldsymbol{A} \sqsubseteq \boldsymbol{B}$ where $\boldsymbol{S}=\{a,b\}$, is ______

3. (10 points) Consider the following argument:

Each student is either male or female.

A student is male if and only if he wears glasses.

Not all students wear glasses.

Then some students are female.

Let the domain consist of all students in some class, $F(x)$: "x is female;", $M(x)$ "x is male;" and $G(x)$: "x wears glasses."

1) Write the formal form of the above argument.

2) Determine whether this argument form is valid and justify your answer.

4. (10 points) Suppose that f is a function from A to B, where A and B are finite nonempty sets. Show that $|f(S)| = |S|$ for all subsets S of A **if and only if** f is one-to-one.

5. (10 points) Use the ***well-ordering property*** to show that "every integer greater than 1 has at least one prime divisor."

points) Let $s_1, s_2 \ldots, s_{101}$ be 101 bit strings of length at most 9. Prove that there exist rings, s_i and s_j, where $i \neq j$, that contain the same number of 0's and the same number of or example, strings 001001 and 101000 contain the same number of 0's and the same of 1's.)

7. (10 points) Prove the following identity using a **combinatorial proof**, where n and r are nonnegative integers with $n \geqslant r$.

$$P(n+1,r)(n+1-r) = (n+1)P(n,r)$$

Bonus Question (10 points):

Show that if m and n are integers with $m \geqslant 3$ and $n \geqslant 3$, then $R(m, n) \leqslant R(m, n-1) + R(m-1, n)$. ($R(m, n)$, the Ramsey number, where m and n are positive integers greater than or equal to 2, denotes the minimum number of people at a party such that there are either m mutual friends or n mutual enemies, assuming that every pair of people at the party are friends or enemies.)